

# Pattern Retrieval on Energy Landscapes

From Biological Memory to Artificial Intelligence

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# Part 1

## The Frame

Pattern Retrieval on Energy Landscapes

# From Fragments to Wholes

A single smell can bring back an entire childhood memory  
— the place, the people, the emotion...

Your brain received a **fragment** and reconstructed the **whole**.

**The question:** how associative (or content-addressable) memory works, exactly?

**1982** Hopfield proposes a model of how the brain stores and retrieves memories.

**2017** Vaswani et al. introduce the transformer — the engine behind modern AI.

**2020** Ramsauer et al. prove these are **the same mathematical object**(\*).

# What I hope to cover

- Block 1–2** The classical model: energy, dynamics, Hebbian storage.
- Block 3** Capacity limits and failure modes (AGS phase transition).
- Block 4** Modern Hopfield networks: exponential capacity.
- Block 5** What the theory tells us about AI.
- Block 6** Open directions.

*Interactive simulations will be shown live throughout. (Cross fingers).*

## Part 2

### The Classical Hopfield Network

# Network Architecture

**Definition** (Hopfield, 1982).

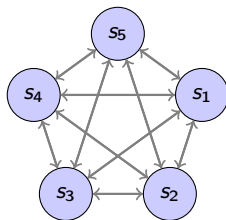
- ▶  $N$  neurons, binary states  $s_i \in \{-1, +1\}$
- ▶ Symmetric weights:  $w_{ij} = w_{ji}$
- ▶ No self-connections:  $w_{ii} = 0$

Network state:  $\mathbf{s}(t) \in \{-1, +1\}^N$

Weight matrix:

$$W = W^T, \quad \text{diag}(W) = 0$$

State space: hypercube  $\{-1, +1\}^N$  with  $2^N$  vertices.



Fully connected  
 $w_{ij} = w_{ji}$

| Neuroscience        | Physics           |
|---------------------|-------------------|
| $s_i = +1$ : firing | Spin up           |
| $s_i = -1$ : silent | Spin down         |
| $w_{ij}$ : synapse  | Coupling $J_{ij}$ |

# Energy Function and Dynamics

**Energy function** (Lyapunov function):

$$E(\mathbf{s}) = -\frac{1}{2} \mathbf{s}^\top \mathbf{W} \mathbf{s} - \boldsymbol{\theta}^\top \mathbf{s}$$

Identical to the Ising spin glass Hamiltonian under  $\sigma_i = s_i$ ,  $J_{ij} = w_{ij}$ ,  $h_i^{\text{ext}} = \theta_i$ .

**Asynchronous update rule:** pick neuron  $i$ , set

$$s_i(t+1) = \text{sign} \left( \sum_{j=1}^N w_{ij} s_j(t) - \theta_i \right)$$

where  $h_i = \sum_j w_{ij} s_j - \theta_i$  is the **local field**.

Each neuron decides using only its own inputs — no global coordinator.

# Convergence

**Theorem** (Hopfield, 1982). Under asynchronous updates with symmetric  $W$  and  $w_{ii} = 0$ :

$$E(\mathbf{s}(t+1)) \leq E(\mathbf{s}(t))$$

with equality iff no neuron changes state.

**Proof idea.** Flipping neuron  $k$  to align with its local field gives

$$\Delta E = - \underbrace{(s_k^{\text{new}} - s_k^{\text{old}})}_{\neq 0} \underbrace{h_k}_{\text{same sign as } s_k^{\text{new}}} \leq 0.$$

Symmetry of  $W$  is essential: it ensures  $h_k$  does not depend on the update order.

**Corollary.** Finite state space ( $2^N$ ) + monotone decrease  $\Rightarrow$  convergence to a **fixed point** in finite time.



## Memory Storage: Hebbian Learning

Store  $M$  binary patterns  $\xi^\mu \in \{-1, +1\}^N$ ,  $\mu = 1, \dots, M$ .

**Hebbian rule** (Hebb, 1949):

$$w_{ij} = \frac{1}{N} \sum_{\mu=1}^M \xi_i^\mu \xi_j^\mu \quad (i \neq j)$$

In matrix form, with  $\Xi = [\xi^1 \mid \dots \mid \xi^M] \in \mathbb{R}^{N \times M}$ :

$$W_{\text{Hebb}} = \frac{1}{N} \Xi \Xi^\top - \frac{M}{N} \mathbf{I}$$

**Biological interpretation:** “Neurons that fire together, wire together.”

$\xi_i^\mu \xi_j^\mu = +1$  (co-active)  $\Rightarrow$  strengthen;  $\xi_i^\mu \xi_j^\mu = -1 \Rightarrow$  weaken.

Model for **Long Term Potentiation/Depression** (Bliss & Lømo, 1973 - Douglas & Goddard, 1975).

The rule is **local**: each synapse uses only its own pre- and post-synaptic activity.

# Pattern Retrieval and the Overlap

**Overlap** (order parameter):

$$m^\mu(\mathbf{s}) = \frac{1}{N} \sum_{i=1}^N s_i \xi_i^\mu = \frac{1}{N} \mathbf{s}^\top \boldsymbol{\xi}^\mu \in [-1, 1]$$

$m^\mu = 1$ : perfect retrieval.  $m^\mu = 0$ : no correlation.  $m^\mu = -1$ : reversed.

**Signal-to-noise decomposition.** Apply  $W_{\text{Hebb}}$  to pattern  $\boldsymbol{\xi}^\nu$ :

$$h_i = \underbrace{\xi_i^\nu}_{\text{signal}} + \underbrace{\sum_{\mu \neq \nu} \xi_i^\mu m^\mu}_{\text{crosstalk noise}}$$

If patterns are orthogonal ( $m^\mu(\boldsymbol{\xi}^\nu) = 0$  for  $\mu \neq \nu$ ): crosstalk vanishes, each  $\boldsymbol{\xi}^\mu$  is a **stable fixed point**.

For random patterns: crosstalk  $\sim \mathcal{N}(0, M/N)$  by CLT. This sets the stage for capacity analysis.

[Live demo: [hopfield\\_simulation.html](http://hopfield_simulation.html)]

# Energy Landscape

The energy in terms of overlaps:

$$E(\mathbf{s}) = -\frac{N}{2} \sum_{\mu=1}^M (m^{\mu})^2 + \frac{M}{2}$$

**Stored patterns** sit at deep energy minima.

**Spurious states** — fixed points not among stored patterns:

1. Mixture states:  $\mathbf{s} \approx \text{sign}(\xi^1 + \xi^2 + \xi^3)$   
moderate overlap with some stored patterns simultaneously
2. Spin-glass states: low overlap with ALL stored patterns
3. Reversed patterns:  $-\xi^{\mu}$

Retrieval = descent into the nearest valley.

**Basin of attraction** of  $\xi^{\mu}$ : all initial states that converge to it.

For low loading ( $\alpha \ll 0.138$ ): basin radius  $\approx 0.15 N$  flipped bits (Amit et al., 1987).

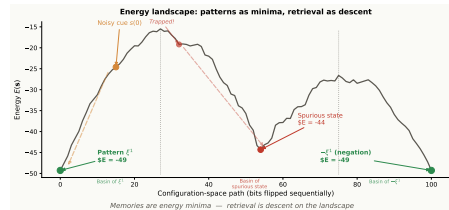
**Key references:**

Hopfield (1982), *PNAS* 79(8)

Hebb (1949), *The Organization of Behavior*

Bliss & Lømo (1973), *J. Physiol.* 232(2)

Amit, Gutfreund & Sompolinsky (1985), *PRL* 55(14)



[figure]

## Part 3

### Capacity and Failure Modes

# The Critical Capacity

Store  $M$  patterns in  $N$  neurons. Define the **loading parameter**:  $\alpha = M/N$ .

**Theorem** (Amit, Gutfreund & Sompolinsky, 1985). For random binary patterns,  $N \rightarrow \infty$ :

$$\alpha_c \approx 0.138$$

Self-consistency equation for the overlap:

$$m = \operatorname{erf}\left(\frac{m}{\sqrt{2\alpha(1-m^2)}}\right)$$

At  $\alpha_c$ , the non-trivial solution ( $m \approx 0.97$ ) vanishes **discontinuously** — a first-order phase transition.

**Rigorous bounds:**

McEliece et al. (1987): exact retrieval for  $M < N/(2 \ln N)$ .

Talagrand (2010): replica-symmetric solution correct for  $\alpha < \alpha_c$ .

**Concrete example:**

$N = 10,000$  neurons  $\Rightarrow$  at most  $\approx 1,400$  memories.

# The Phase Diagram

Three regimes, separated by sharp transitions at  $T = 0$ :

| Phase        | $\alpha$              | Behaviour                                           | Ising analogue |
|--------------|-----------------------|-----------------------------------------------------|----------------|
| Retrieval    | $< 0.138$             | Stored patterns are stable.<br>$m \approx 1$ .      | Ferromagnetic  |
| Spin glass   | $0.138\text{--}0.318$ | Spurious states dominate.<br>$m \approx 0, q > 0$ . | Spin-glass     |
| Paramagnetic | $> 1/\pi$             | No retrieval. $m = 0, q = 0$ .                      | Paramagnetic   |

$m^\mu = \frac{1}{N} \mathbf{s}^\top \boldsymbol{\xi}^\mu$  (overlap) is the order parameter.

$q = q_{EA} = \frac{1}{N} \sum_i \langle s_i \rangle_T^2$  (Edwards–Anderson) distinguishes spin-glass from paramagnetic.

Ref: Amit, Gutfreund & Sompolinsky (1987), *Ann. Phys.* 173(1); Mézard, Parisi & Virasoro (1987).

# Spurious States: Two Readings of Failure

## Types:

### 1. Mixture states:

$$\text{sign}(\xi^1 + \xi^2 + \xi^3)$$

### 2. Spin-glass states:

Low overlap with all patterns.

### 3. Reversed patterns:

$-\xi^\mu$ , always a fixed point.

| Phenomenon      | Neuroscience    | AI                 |
|-----------------|-----------------|--------------------|
| Mixture         | Memory blend    | Hallucination      |
| Spin-glass      | Confabulation   | Confident nonsense |
| Near $\alpha_c$ | Memory overload | Context saturation |

Spurious states are **mathematical necessities** of finite-capacity landscapes



# The Hard Ceiling

Linear capacity: at most  $\sim 0.14 N$  patterns in  $N$  neurons.

| System        | $N$            | $M_{\max} \approx 0.14 N$ |
|---------------|----------------|---------------------------|
| Small network | $10^3$         | $\sim 140$                |
| Human cortex  | $\sim 10^{11}$ | $\sim 10^{10}$            |

Three bottlenecks: **quadratic** energy, **binary** states, **linear** signal-to-noise scaling.

**What if you could break through the linear barrier?**

[Live demo: [hopfield\\_capacity.html](http://hopfield_capacity.html)]

## Part 4

The Modern Hopfield Network  
and the Attention Equivalence

# Breaking Through the Ceiling

|             | Classical (1982)                                               | Modern (2016/2020)                    |
|-------------|----------------------------------------------------------------|---------------------------------------|
| State space | $\{-1, +1\}^N$                                                 | $\mathbb{R}^d$                        |
| Energy      | Quadratic: $-\frac{1}{2}\mathbf{s}^\top \mathbf{W} \mathbf{s}$ | Log-sum-exp                           |
| Interaction | Pairwise ( $n = 2$ )                                           | All orders ( $n \rightarrow \infty$ ) |
| Update      | $\text{sign}(h_i)$ , iterative                                 | Softmax, <b>one step</b>              |
| Capacity    | $M \sim 0.14 N$                                                | $M \sim e^{d/2}$                      |

From **linear** to **exponential**. Not an incremental improvement — a qualitative leap.

Ref: Krotov & Hopfield (2016), *PNAS* 113(9); Ramsauer et al. (2020), arXiv:2008.02217.

# The Modern Hopfield Network

Store  $M$  patterns as columns of  $X = [\xi^1 \mid \dots \mid \xi^M] \in \mathbb{R}^{d \times M}$ . Query:  $\xi \in \mathbb{R}^d$ .

**Energy:**

$$E(\xi) = - \underbrace{\text{lse}(\beta X^\top \xi)}_{\text{pattern similarity}} + \underbrace{\frac{1}{2} \|\xi\|^2}_{\text{regularization}} + \text{const}, \quad \text{lse}(z) = \frac{1}{\beta} \log \sum_{\mu} e^{\beta z_{\mu}}$$

**Update rule** (one step):

$$\xi^{\text{new}} = X \text{ softmax}(\beta X^\top \xi)$$

**Capacity theorem** (Ramsauer et al., 2020):

$$M = e^{d/2}, \quad \text{retrieval error } \epsilon = O(e^{-d})$$

Requires a minimum **overlap gap** between self-overlap and cross-overlap (see Block 5).

# The Krotov–Hopfield Interpolation

The modern construction is the endpoint of a family parametrized by interaction order  $n$ :

$$E_n(\mathbf{s}) = - \sum_{\mu=1}^M F_n \left( \sum_i \xi_i^\mu s_i \right), \quad F_n(x) = x^n$$

| Interaction order      | Energy      | Capacity                  |
|------------------------|-------------|---------------------------|
| $n = 2$ (classical)    | Quadratic   | $M \sim 0.14 N$           |
| $n = 3$                | Cubic       | $M \sim \alpha_3 N^2$     |
| $n \geq 3$             | Polynomial  | $M \sim \alpha_n N^{n-1}$ |
| $n \rightarrow \infty$ | Log-sum-exp | $M \sim e^{\alpha N}$     |

Each step in  $n$  multiplicatively improves capacity.

Ref: Krotov & Hopfield (2016), *PNAS* 113(9); Krotov & Hopfield (2021), *ICLR*.

# Hopfield Retrieval = Transformer Attention

**Modern Hopfield update:**

$$\xi^{\text{new}} = X \text{ softmax}(\beta X^T \xi)$$

**Scaled dot-product attention** (Vaswani et al., 2017):

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

**The dictionary:**

|                        |         |                  |
|------------------------|---------|------------------|
| Query                  | $\xi^T$ | $= Q$            |
| Stored patterns (keys) | $X^T$   | $= K = V$        |
| Inverse temperature    | $\beta$ | $= 1/\sqrt{d_k}$ |

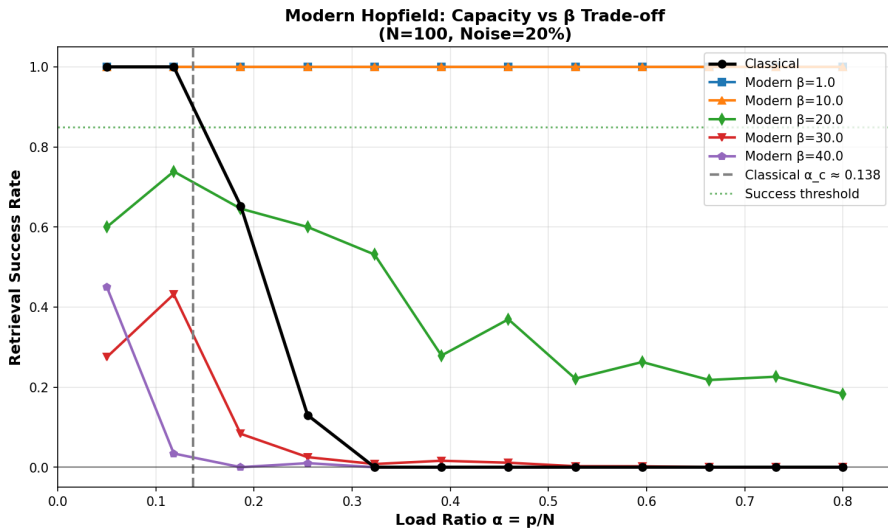
**These are the same equation.**

Hetero-associative extension ( $K \neq V$ ):  $\xi^{\text{new}} = V \text{ softmax}(\beta K^T \xi)$ .

Ref: Ramsauer et al. (2020); Vaswani et al. (2017), *NeurIPS*.

[Live demo: [modern\\_hopfield\\_beta.html](https://modern-hopfield-beta.html)]

# Modern Hopfield: Capacity vs $\beta$ Trade-off



# A Physics of Attention

Because Hopfield retrieval = attention, the theoretical apparatus transfers:

| Hopfield theory                    | Consequence for transformers                                   |
|------------------------------------|----------------------------------------------------------------|
| Energy landscape                   | Each attention head performs energy minimisation               |
| Capacity $M = e^{d/2}$             | Exponential storage; requires pattern separation               |
| Three fixed-point types            | Precise retrieval, blended output, or degenerate averaging     |
| Temperature $\beta = 1/\sqrt{d_k}$ | LLM temperature tunes retrieval sharpness                      |
| Spurious states                    | Hallucinations = convergence to spurious attractors            |
| Multi-head attention               | $N$ parallel Hopfield networks, each with capacity $e^{d_k/2}$ |

For the physicists: the most commercially important computation on the planet is **energy minimisation**.

For the data scientists: you now have a **physics of attention**.



## Part 5

What the Theory Tells Us About AI

## Prediction 1: Three Modes of Attention

Every attention head is a Hopfield network. Its fixed points classify into three types:

| Mode       | Softmax weights                    | Output                 | Example                                  |
|------------|------------------------------------|------------------------|------------------------------------------|
| Pure state | $p_\mu \approx 1$ for one $\mu$    | Sharp, specific        | “Capital of France?”                     |
| Mixture    | $p_\mu \approx 1/k$ over $k$ items | Blended, creative      | Similar meetings $\rightarrow$ composite |
| Average    | $p_\mu \approx 1/M$                | Diffuse, uninformative | Long context $\rightarrow$ noise         |

The LLM temperature  $T = 1/\beta$  moves the system across these regimes:

Low  $T$ : pure states (precise, repetitive)  $\longleftrightarrow$  High  $T$ : averages (random, incoherent)

The “creative zone” is the metastable mixture regime. This is a consequence of the phase structure.

## Prediction 2: The Overlap Gap Controls Retrieval

When does attention succeed? When the pattern's **self-overlap** dominates every **cross-overlap**.

**Self-overlap** of pattern  $\mu$ :

$$q_{\text{self}}^{\mu} = (\xi^{\mu})^{\top} \xi^{\mu} = \|\xi^{\mu}\|^2$$

The inner product of a pattern with itself—how strongly it activates its own retrieval.

**Cross-overlap** with pattern  $\nu$ :

$$q_{\text{cross}}^{\nu\mu} = (\xi^{\nu})^{\top} \xi^{\mu}$$

The inner product between different patterns—how much  $\nu$  interferes with retrieval of  $\mu$ .

Define the **overlap gap**:  $\Delta^{\mu} = q_{\text{self}}^{\mu} - \max_{\nu \neq \mu} q_{\text{cross}}^{\nu\mu}$

**Separation condition** (Ramsauer et al., 2020):

$$\Delta^{\mu} > \frac{1}{\beta} \left( \log(M-1) + \log \frac{1}{\epsilon} \right)$$

The gap must exceed a threshold that grows only **logarithmically** in  $M$ . For random patterns on the sphere,  $q_{\text{self}} = d$  while  $q_{\text{cross}} \sim \sqrt{d}$ , so the gap grows linearly—this is why  $M = e^{d/2}$  is achievable.

## When the Overlap Gap Closes: Hallucination

When similar items crowd the context, cross-overlaps approach the self-overlap, and  $\Delta^\mu$  shrinks below threshold.

**Hallucination** = convergence to a spurious attractor: a fixed point that does not correspond to any stored pattern.

| Mechanism         | Overlap-gap diagnosis                                     | Intervention                                         |
|-------------------|-----------------------------------------------------------|------------------------------------------------------|
| Mixture           | $\Delta^\mu$ small: cross-overlaps $\approx$ self-overlap | Lower $T$ ; RAG with diverse documents; better $W^K$ |
| Spin-glass        | Frustrated landscape: no single pattern dominates         | Fine-tuning (RLHF/DPO); increase $d$                 |
| Capacity overflow | $M > e^{\beta d/2}$ : too many patterns                   | Shorten context; increase $d$ ; sparse attention     |

The theory classifies hallucination causes and predicts that **different types need different fixes**.

Ref: Ramsauer et al. (2020); Xiao et al. (2024), *ICLR*.

## Prediction 3: Depth = Number of Retrieval Steps

Each transformer layer performs one Hopfield retrieval step on its own energy landscape.

| Hopfield theory                                      | Transformer implication                                                        |
|------------------------------------------------------|--------------------------------------------------------------------------------|
| One step $\rightarrow$ pure-state retrieval          | Factual recall, entity recognition: early layers suffice                       |
| Multiple steps $\rightarrow$ distant query converges | Multi-hop reasoning: each hop is a layer                                       |
| Different landscape per layer                        | Hierarchical features: syntax $\rightarrow$ semantics $\rightarrow$ pragmatics |
| Convergence $\rightarrow$ fixed point                | Layer saturation: later layers become near-identity                            |

**Chain-of-thought** converts depth into context length: intermediate results are externalised as new stored patterns, enabling single-hop retrieval at each step.

**Residual stream:**  $\xi^{(\ell+1)} = \xi^{(\ell)} + \Delta^{(\ell)}$ . Each layer nudges the representation toward an attractor.

Ref: Wei et al. (2022), *NeurIPS*.

# What Hopfield Theory Can and Cannot Do for AI

## What it delivers (✓)

- ✓ An **energy function** for every attention head
- ✓ **Capacity**:  $M = e^{d/2}$  (\*), with computable overlap-gap condition
- ✓ **Three output modes** classified by fixed-point type, controlled by  $\beta$
- ✓ **Hallucination taxonomy**: three mechanisms, three interventions
- ✓ **Convergence**: dynamics always decrease energy
- ✓ Principled **RAG** design (maximise overlap gap) and **temperature tuning**

## Where it stops (✗)

- ✗ **Training**: inference-time theory only; how  $W^Q, W^K, W^V$  are learned is outside scope
- ✗ **Feedforward layers**: only attention is modelled; MLPs are separate
- ✗ **Cross-layer dynamics**: each layer analysed in isolation; composed system is open
- ✗ **Specific tokens**: predicts retrieval *modes*, not *which* tokens appear
- ✗ **Structured data**: assumes random patterns; real data has correlations

The only theory of attention with rigorous capacity theorems and phase transitions—but a **single-layer, inference-time** theory, not a complete model of the transformer.

## Part 6

Open Directions

# What's Next: Four Open Questions

## 1. Can we predict when AI systems fail?

The overlap gap  $\Delta$  is computable on trained models. Can it predict which queries will hallucinate *before* running inference?

## 2. Where do scaling laws come from?

Capacity  $M = e^{d/2}$  gives a dimension–capacity relationship. Can this derive the empirical scaling exponents (Kaplan et al., 2020)?

## 3. What happens with real (non-random) data?

The theory assumes i.i.d. patterns. How does the phase diagram deform for correlated data?

## 4. Is attention a universal computation?

Hopfield networks appear in neuroscience, AI, biology, and economics. Is energy-based retrieval a universal motif?



What systems in your field have this structure?

# Key References

## Foundational:

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Krotov & Hopfield (2021) [[arxiv](#)]

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